

Neuro-Symbolic Agents for Safe, Sovereign, and Auditable AI Systems on the Shop Floor

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Abstract

Industrial and manufacturing settings are on the verge of a **physical AI** wave. Agents are expected to maintain plants, collaborate with humans on shop floors, and act autonomously in regulated facilities. While today’s agentic AI stacks are maturing rapidly, they stall at the factory gate. We argue that the issue lies not in the models themselves, but in the missing substrate: machine-verifiable guarantees of **safety** and **sovereignty** that operators, safety engineers, and auditors require before allowing any agent to interact with an asset. Furthermore, we argue that the agents fit for the shop floor are not monolithic LLMs, but rather, **neuro-symbolic agents**. An LLM proposes while established, verifiable AI methods (e.g., automated planning, constraint solving, and formal verification) and physics-based **simulations** dispose. Any action on the physical world must pass through a simulation gate before being released. These agents *learn* by deriving executable skills from experience, but only within a bounded, human-controlled framework that never unilaterally expands their safety envelope. We analyze why the latest generation of personal agents and emerging agent-interoperability standards do not meet these expectations. We also propose a protocol-first **substrate** that ensures the safety and sovereignty of such agents while disciplining their interactions with one another and with humans. Our proposal is based on two ongoing projects of ours, the Soofi and LxM research projects, but we present it as platform-independent concepts. The paper concludes with specific items for the Trustworthy Agentic AI Roadmap.

1 Physical AI Meets the Shop Floor

AI is key to the solutions needed for the global success of Industry 4.0, which we launched in Germany in 2011. However, generative AI alone is insufficient for the second decade of Industry 4.0: physical and agentic AI are now at the core of a new wave of innovation based on industrial AI [Kagermann and Wahlster, 2022]. Physical AI brings the promise

of agents that not only reason and converse but also perceive and act in the physical world [Zitkovich *et al.*, 2023; Driess *et al.*, 2023; Bjorck *et al.*, 2025]. The same wave is now reaching industry: vendors propose maintenance copilots, embodied co-workers in human–robot collaboration (HRC) cells, automated-guided-vehicle (AGV) fleet supervisors, and autonomous process optimizers. These scenarios are not speculative; proof-of-concept deployments exist for each one.

Yet the shop floor is not adopting them. Consider the following generic scenario: a fault-prediction agent needs to access PLC (programmable logic controller) signals, propose a maintenance window, and, at a later stage, actuate a valve sequence under operator supervision. Three specific issues block roll-out:

Safety. Industrial machinery is governed by harmonized safety standards (ISO 12100 [ISO, 2010], ISO 10218-1:2025 [ISO, 2025], IEC 61508/62061 [IEC, 2010; IEC, 2021]) and, for AI specifically, by the EU AI Act (Regulation 2024/1689) whose Article 9 mandates risk management for high-risk systems [European Parliament and Council of the European Union, 2024]. Who signs the resulting risk assessment for an *agent*? More importantly: how is it bound to the agent in a machine-interpretable way that a notified body can later verify?

Sovereignty. Plants do not allow operational data to leave the premises. They cannot commit to a single hyperscaler’s model. They reject SDK lock-in that makes the production environment hostage to one vendor’s roadmap. This concern is reinforced by the NIS2 directive (the EU’s second network-and-information-security directive), the General Data Protection Regulation (GDPR), and the European data-sovereignty agenda (the Manufacturing-X and Gaia-X initiatives).

Auditability. Every agent action must be reconstructable weeks later, in front of safety inspectors, notified bodies, or internal audit, including the LLM call that produced a decision, the tool result that confirmed it, and the human signature that authorized it.

Today’s agent stacks address none of these three concerns. The missing element is not yet another framework but a *substrate*: a small set of verifiable bindings that turn safety—and, beyond it, regulatory compliance at large—together with sovereignty and auditability from policy statements into

machine-checkable contracts. The remainder of the paper lays out the three axes of the industrial design space (Section 2) and analyzes why the current state of the art falls short on all three (Section 3). It then makes safety verifiable (Section 4), develops the neuro-symbolic agent model and its interaction patterns (Sections 5–6), and contributes concrete roadmap items for the SAFER-2026 Trustworthy Agentic AI Roadmap (Section 7).

2 Design Space for Industrial Agents

Industrial agents do not live in the same design space as personal agents. We propose three orthogonal axes that together capture the relevant constraints, then ground them in the two on-going projects behind this paper and in one running example that we thread through the rest of the text.

Safety—environment zones (E1–E4). The first axis asks what it costs to be wrong. An agent’s task says little about the guarantees it must offer; its *environment* does. The same planning capability that is unremarkable at a desk is unacceptable for a moving robot. We therefore order industrial contexts by the physical and regulatory consequence of error, which in turn determines how much verification the substrate must supply. E1 covers benign office and knowledge-work contexts. E2 captures laboratory and prototyping environments. E3 designates regulated zones such as ATEX (explosive-atmosphere) areas, clean rooms, good-manufacturing-practice (GMP) pharmaceutical production, and clinical contexts, where conformance is a legal precondition for operating at all. E4 designates safety-critical zones—safety-integrity-level (SIL) ≥ 2 functional-safety contexts, human–robot collaboration cells, and cooperative autonomous vehicles—where a single wrong actuation can cause injury. The verification burden rises monotonically along the axis, and E3 and E4 are precisely where today’s agent research has produced no usable substrate.

Latency—latency zones (L0–L4). The second axis asks how fast a decision must be made, and it is orthogonal to safety: it constrains not how careful an agent must be but *which kind of agent can act at all*. The scale is anchored at zero by a layer that is deliberately out of scope. L0 is hard real-time (sub-millisecond to millisecond)—safety PLCs, robot inner-loop control, emergency-stop chains—realized in certified, hardwired safety-instrumented systems that no agent, however verifiable, may enter. We refer to it as 0 because it is the foundation upon which the agents are built and which they must adhere to, rather than a range that they occupy. Agentic deployment begins one zone up. L1 is soft real-time (tens of milliseconds): regulation, sensor fusion, collision avoidance. L2 is interactive (around 100 milliseconds): human–machine dialog, manufacturing-execution-system (MES) reactions, edge inference. L3 is workflow-scale (seconds to hours): dispatching, maintenance planning, multi-agent negotiation. L4 is reflexive (hours to days): learning loops, skill promotion, audit synthesis. The further up this axis a task sits, the more deliberation it can afford, and only above L1 does an LLM in the loop become physically admissible.

Sovereignty—four dimensions. The third axis asks who keeps control once an agent is deployed. For an industrial operator this is not a preference but a condition of adoption: production data, the models that read it, the protocols that carry decisions, and the authority to act must not silently migrate to a platform vendor. Sovereignty is what keeps a capable agent from becoming locked into external dependence, and it can be broken down into four dimensions, each of which can be considered independently of the others. *Data sovereignty*: operational data and agent memory remain on premises; the agent’s private, append-only episode record is the audit substrate. *Model sovereignty*: the LLM provider is exchangeable—local open-weight, sovereign-cloud, or hyperscaler—without changing the agent’s bindings, and the evaluation and fine-tuning datasets distilled from operational experience stay behind the operator’s firewall, with recording an episode and releasing it for training as two separate approvals. *Protocol sovereignty*: open standards—the Agent-to-Agent protocol (A2A), the CloudEvents event specification, the Asset Administration Shell (AAS), and the Model Context Protocol (MCP)—replace proprietary SDKs. *Operational sovereignty*: the operator and the safety engineer sign deployments, not the platform vendor.

Two running research projects. The position is grounded in two ongoing research projects by the authors; the rest of the paper then abstracts from them to the underlying, platform-independent concepts. **LxM** is a research project at DFKI for the compliant, (semi-)autonomous planning and execution of business processes within and, increasingly, across enterprise boundaries, with the human in the role of decision-maker. *LxM* is our umbrella term for the specialized derivatives of large language models it combines—large language models (LLM), large reasoning models (LRM), large action models (LAM) [Zhang *et al.*, 2025], and large process models (LPM) [Kampik *et al.*, 2024]—fused, through neuro-symbolic methods, with an *integrated information architecture* over standardized digital twins (AAS), OWL-based knowledge graphs, and language models. It is evaluated along three make-to-order use cases: a multi-agent supply chain with transitive requirement catalogs; quality-aware, safe production in DFKI’s MRK4.0 human–robot-collaboration lab (an E4 setting); and compliance process monitoring in a regulated industry (an E3 setting). The **Soofi** project aims on the one hand at training a family of sovereign open-source foundation models. On the other it develops a protocol-first reference platform on which industrial agents run: verifiable contracts are fixed at a *protocol* layer and mapped onto exchangeable binding and reference layers; each agent keeps a private, append-only record of its episodes, and a shared layer distills cross-agent knowledge and skills into common semantic memory. We draw on both efforts for concrete grounding, including the running example, but everything that follows is stated as a concept, not a platform feature.

A running example. To keep the discussion concrete, we thread one example through the paper, drawn from discrete manufacturing. On a powertrain-component line, a CNC machining cell drifts slowly over weeks—spindle vibration creeps up, tool life shortens—and it appears that the devia-

tion correlates with a batch of forged blanks that has a wider range of hardness values. Three heterogeneous agents share the line but no codebase: an *edge* agent on the cell controller, an *engineering* agent on the workstation, and a *supply-chain* agent in the site cluster. We return to it to make each axis and mechanism tangible.

A central implication. Not every industrial agent is an LLM agent. At L1—the lowest band agents occupy, just above the certified L0 floor—latency and verifiability constraints, together with functional-safety norms such as IEC 61508 and IEC 62061, make LLMs unusable in the inner loop. This layer requires *rule-based, formally verifiable agents* that inherit directly from the multi-agent-systems (MAS) legacy of Section 3.1, from IEC 61499 function-block control, and from deterministic policy architectures. Neuro-symbolic agents with an LLM at their core only begin to dominate at L2 and above. Even there, the neural core does not need to be large: small language, action, and process models are gaining popularity for industrial AI because they are generally faster, cost less, consume less power, and are easier to manage than their large counterparts [Wahlster, 2026]. Focused on a single domain, they are less prone to hallucinations and to highly ambiguous responses—and some ultra-compact models challenge the core assumption driving today’s AI arms race, that bigger is always better, by outperforming LLMs on various benchmarks. Industrial agent systems are therefore inherently *heterogeneous*, and any candidate substrate must support heterogeneous agent types, not only LLM wrappers. Coordinating these heterogeneous types—rule-based controllers, LLM-driven planners, and humans—under one shared contract is itself a core concern, taken up in Section 6; any candidate substrate must be shaped to carry all of them. The running example spans the whole latency axis at once: the edge agent runs at L1–L2 (certifiable, LLM-free), the engineering agent at L3 (drafting a feed-reduction patch for the cell’s PLC runtime, TwinCAT, as a reviewed pull request), and the supply-chain agent at L4—three implementations (Rust, TypeScript, Python) that share only the protocol.

3 State of the Art

The relevant previous work falls into four parts: an industrial multi-agent-systems legacy (Section 3.1) largely forgotten by today’s agent community; a current landscape (Section 3.2) shaped by foundation-model robotics and shifting industrial norms; a dominant class of personal agents (Section 3.3) demonstrably unfit for the shop floor; and agent-interoperability standards (Section 3.4) that solve coordination but not safety or sovereignty.

3.1 Multi-Agent Systems Before the LLM Era

MAS research has a thirty-year industrial heritage that is rarely cited by current agentic-AI work. The BDI (belief–desire–intention) architecture provided a deliberative agent model with explicit goals and plans [Rao and Georgeff, 1991; Rao and Georgeff, 1995]. Agent communication was formalized through the FIPA Agent Communication Language (FIPA-ACL) [FIPA, 2002] and the earlier KQML, with the Contract Net Protocol as a canonical coordination primitive

[Smith, 1980]. The JADE platform [Bellifemine *et al.*, 2007], Jason and Jadex provided implementations that saw real industrial use.

Crucially, several of these efforts targeted production directly. The PROSA reference architecture for Holonic Manufacturing Systems [Van Brussel *et al.*, 1998] decomposed manufacturing control into resource, product, and order holons. ADACOR and ADACOR² [Leitão and Restivo, 2006; Barbosa *et al.*, 2015] introduced adaptive coordination layers, and IEC 61499 [IEC, 2012] with agent extensions opened the door to function-block-based agentic control with formal real-time properties.

This generation achieved formal communication semantics, negotiation patterns, shared ontologies, and decentralized decision models, but failed to scale: perception, language, and learning were three decades away, engineering overhead was prohibitive, and service-oriented architectures absorbed the attention [Leitão *et al.*, 2016; Karnouskos *et al.*, 2020]. Today’s LLM-based stacks solve the learning side but ignore this body of work—yet its communication semantics, negotiation patterns, and industrial-coupling techniques are exactly what current agentic AI lacks.

Indeed, the very term *agent* has shifted meaning. In the MAS tradition an agent was a persistent, autonomous entity with its own goals, beliefs, and the social ability to negotiate—an actor with an agenda [Wooldridge and Jennings, 1995]. Today the word most often denotes an *agent loop*: a language model invoking tools, observing results, and re-planning in a ReAct-style cycle [Yao *et al.*, 2023]. The shift bought generality and language at the cost of the very constructs—explicit authority, goals, and negotiated commitments—on which industrial coordination depends.

3.2 Current Landscape

Two contemporary developments matter for our argument. First, the physical-AI wave has produced vision–language–action (VLA) models that act in the physical world, from RT-2 [Zitkovich *et al.*, 2023] and PaLM-E [Driess *et al.*, 2023] to NVIDIA’s GR00T N1 [Bjorck *et al.*, 2025] and Google DeepMind’s Gemini Robotics [Gemini Robotics Team *et al.*, 2025]. These advances are real but expose a widening safety gap—perception–action loops on foundation models have no native concept of an environment zone (collaborative cell vs. clean room vs. ATEX area) and offer no verifiable risk binding.

Second, industrial safety norms have shifted. ISO 10218-1:2025 [ISO, 2025] absorbs the previous technical specification on collaborative robots (ISO/TS 15066:2016) and introduces, for the first time, explicit cybersecurity requirements for industrial robots—a normative anchor point that did not exist at the start of the agentic-AI wave. Both axes have moved, then: model capability now reaches into the physical world, and the normative frame has begun to acknowledge it. What has *not* moved is the layer in between—a machine-verifiable substrate that would let either advance be deployed under explicit safety and sovereignty guarantees. The remainder of this section shows that none of the agent classes and standards in active use supplies it.

3.3 Personal Agents—Built for Individuals, Not for Plants

Most attention has recently flowed to a class of agents that ship with extraordinary capability but a deeply individualist design: Claude Code [Anthropic, 2024a], AutoGPT [Significant Gravititas, 2023], the Letta/MemGPT family [Packer *et al.*, 2024], Manus, Devin, Cursor Agents, and OpenAI’s Operator and Computer-Use products. These systems excel at coding and knowledge work, support per-user persistent memory, and call tools via MCP-style protocols. A still newer, virally popular wave is explicitly local-first: OpenClaw [Steinberger and The OpenClaw Project, 2025] and the Hermes Agent [Nous Research, 2026] run as self-hosted background processes, keep memory as plain on-disk files, and distill completed tasks into reusable skills—coming closer to operational sovereignty and self-improvement than any cloud product, yet sharing the same individualist design.

Four properties make them unfit for the shop floor. (1) *Single-stakeholder model*: every artifact (memory store, tool credential, audit trail) is owned by one user; there is no separation between safety engineer, operations manager, agent runtime, and auditor—a separation industrial deployments require by both convention and law. (2) *Sovereignty is configurable, not constructed*: harnesses such as Claude Code and opencode can be pointed at self-hosted models, and OpenClaw and Hermes are local-first by design—but this is an opt-in deployment choice, not a property the agent can present and an operator can verify, and the surrounding stack (tool credentials, telemetry, audit sinks) still typically assumes cloud, conflicting with NIS2, GDPR, and in-house operational-data policies. (3) *No verifiable safety context*: nothing in the agent’s representation distinguishes an ATEX-Zone-2 task from a clean-room task from a benign office task. (4) *Audit trails are absent or proprietary*: traces are stored as opaque vendor logs, not as signed, hash-chained, machine-interpretable episodes that survive a vendor change and enable third-party reproduction.

The asymmetry is striking: these stacks are an order of magnitude more capable than the MAS generation of Section 3.1 on perception and language, yet an order of magnitude less suited to industrial deployment on every axis that matters to an operator.

3.4 Agent Standards—Interoperability $\neq$ Safety $\neq$ Sovereignty

Agent-interoperability standards have consolidated rapidly: Anthropic’s MCP [Anthropic, 2024b], Google’s A2A [Linux Foundation, 2026], and Cisco-led AGNTCY [AGNTCY Project, 2025] all moved into the *Agentic AI Foundation* under the Linux Foundation between mid- and end-2025 [Anthropic, 2025], giving agentic standards a single neutral steward for the first time. Yet they barely compete: they occupy three orthogonal layers. *Vertically*, an agent reaches down to tools, APIs, and data sources: the province of MCP. *Horizontally*, agents delegate and coordinate across organizational boundaries: the province of A2A. Above both sits an *infrastructure* layer—discovery, identity, messaging, observability—the province of AGNTCY and its Open Agentic Schema Framework (OASF). The crucial observation for

this paper is that all three layers concern *coordination*; none of them, nor the process-level governance frameworks below, binds a safety or a sovereignty contract.

MCP solves tool invocation and result handling; it deliberately leaves tool authentication, provenance, and safety semantics open. A2A solves discoverability via signed Agent Cards that advertise an agent’s *capabilities*—its skills and interaction modes; but capability is not authority, and the card constrains neither what an agent is *authorized* to do, nor where, nor under whose risk assessment. AGNTCY provides ecosystem-level coordination but has no industrial safety layer. The AAS (IEC 63278 [IEC, 2023]) together with IDTA’s submodel catalog is the most mature substrate for industrial-asset semantics and data sovereignty—yet it embodies the single most concrete substrate gap. Despite active work at the Industrial Digital Twin Association (IDTA), no normative AAS submodel for agentic actors exists at the time of writing: only submodels for static AI *models* (AIModelNameplate) and datasets (AIDataset) have been published [IDTA, 2025]. Agents—which are stateful, evolve through learning, and operate under negotiable authority—fit none of these. On the governance side, the NIST AI Risk Management Framework (AI RMF) [Tabassi, 2023], ISO/IEC 42001:2023 [ISO/IEC, 2023b] and ISO/IEC 23894:2023 [ISO/IEC, 2023a], together with EU AI Act Article 9 [European Parliament and Council of the European Union, 2024], mandate risk management at a process level—but produce no machine-interpretable artifacts that an agent can carry, present, and have verified at runtime.

Conclusion of Section 3. Three worlds exist in parallel: a forgotten **MAS legacy with industrial depth but without LLMs**; an agile **agent-stack and standards world without industrial depth**; a mature **industrial-standards world without agent semantics**. Nobody is building the bridge between them. **We propose exactly that bridge—and argue it must be a verifiable substrate, not another framework.**

4 Making Safety Verifiable

A viable substrate turns safety from a documentary practice into a set of machine-checkable contracts. One prerequisite runs through all of them: every agent—not only the physical assets it works with—must itself be a first-class industrial asset described by its own AAS, closing the submodel gap diagnosed in Section 3.4; this shell is where the contracts below attach. Five mechanisms carry this, framed from the operator’s point of view.

(1) Risk assessment becomes a machine-interpretable contract. Every agent instance carries, as part of its AAS, a `RiskAssessment` submodel element signed by the responsible safety engineer with a strong identity—a cryptographic key bound to a person, not an organization. The signature carries a validity period and a digest of the risk-assessment document. Before any action in E3 or E4, the runtime verifies that the agent’s type-level constraints contain the instance-level assessment, that the signature is valid, that the validity window covers the current time, and that the digest matches: the runtime re-hashes the risk-assessment document the submodel references and compares the result against the

signed digest, so an edited or substituted document breaks the binding mechanically, with no human in the loop. Episodes recorded without a reference to a valid risk-assessment object are quarantined and never enter the shared cross-agent memory. The risk assessment is deliberately only the sharpest instance of a general pattern: *policy* $\rightarrow$ *machine-interpretable condition* $\rightarrow$ *runtime check*. Regulations, company policies, and customer requirement catalogs arrive as documents referenced from the same AAS; semi-automated knowledge extraction turns their natural-language clauses into formalized conditions over an OWL knowledge graph derived from the AAS content—the integrated information architecture introduced with LxM in Section 2—so that quality obligations (a certification mandating a step sequence and its documentation), data-governance invariants, and supply-chain requirements are checked by exactly the machinery that checks the risk assessment.

(2) New skills cross a risk-review gate. Agents learn—that is, after all, the entire reason to deploy them. But skills whose risk-relevance flag is *envelope-extending* or *envelope-altering* cannot be acted autonomously. They are routed to a pending-risk-review state and are only activated by a signed clause in the agent’s instance-level risk assessment, again signed by the safety engineer.

(3) Every action is auditable through an immutable episode chain. Agent episodes form a hash-linked chain (each episode carries the hash of its predecessor in the same chain), making reordering and silent rewriting detectable. Each episode also carries the OpenTelemetry trace identifier of the underlying runtime activity, so the agent’s *subjective* record (intentions, uncertainty, decision rationale) can be cross-checked against the *objective* telemetry (tool calls, latencies, errors). Discrepancies—for example, an agent reporting success while telemetry shows three failed tool spans—are surfaced to the shared learning layer and influence skill confidence. Audit completeness here means a complete *evidence chain*, not unfiltered capture: depending on the data class, an episode stores content, redacted content, or only hashes and references to authoritative records, so the same chain that satisfies an auditor also respects data-protection and trade-secret constraints.

(4) Authority is separated by role and by key. Three distinct cryptographic keys sign three distinct artifacts: the *safety engineer* signs the RiskAssessment, the *operations manager* signs the OperationsEnvelope (where, when, and under what scope the agent is allowed to act), and the *agent itself* signs its runtime status and episode chain. A per-site revocation list closes the loop. These authorities meet at runtime in an explicit policy layer: every tool call and external effect passes an enforcement point whose outcome—allow, deny, dry-run only, or defer—is decided separately from the human approval it may additionally require (none, single, or dual-control), and a pending approval never acts. No single party—and explicitly not the platform vendor—can unilaterally authorize an unsafe deployment.

(5) Physical actions pass a simulation gate. For any action that touches the physical world in E3 or E4, the agent never

acts directly on its LLM-proposed plan. The plan is first validated in a process- and context-specific, physics-based simulation of the cell—the same digital twin the AAS already describes—and only a plan that passes (no collision, force and speed limits respected, sequence and quality constraints satisfied) is released for real execution. The simulator’s identity, its inputs, and its verdict are recorded in the episode, so an auditor can replay the gate later. The gate operationalizes the neuro-symbolic principle of Section 5: the LLM may propose, but an established, verifiable method—here a simulator—disposes before anything moves.

5 Neuro-Symbolic Agents: Methods, Skills, and Learning

Having made the substrate’s guarantees checkable, we can say what runs on it. The agents are *neuro-symbolic*: a language model proposes, verifiable methods and physics-based simulation dispose. We follow that division from the inside out—the methods that make a single decision defensible, the skills that turn recurring decisions into cheap routine, and the bounded learning loop that lets both improve without escaping the safety envelope of Section 4.

Methods: the LLM proposes, verifiable methods dispose.

The agents we consider are not monolithic LLMs. Internally, each is organized as a *holon* in the manufacturing-control tradition of Section 3.1: a coordinating core delegates to task-specific expert agents and to agentic tools for data access, logical planning, context query, scheduling, and simulation—the holonic decomposition of PROSA re-emerging one level down, inside the agent. Following the fast/slow distinction of human cognition [Kahneman, 2011], an LLM supplies fast, intuitive proposals, while established, *verifiable* AI methods—automated planning over PDDL (Planning Domain Definition Language) domains [Ghallab *et al.*, 2004], constraint solving, and formal verification—together with physics-based simulation supply the slow, checkable judgment [Garcez and Lamb, 2023]. Every such call—solver, planner, simulator—is a first-class, auditable tool invocation recorded in the agent’s episode. This is what makes the agent defensible to a safety engineer: the trustworthiness of a safety-critical decision rests on a certifiable, decades-mature method, not on the statistical behavior of a language model. The simulation gate of Section 4 enforces this delegation before any physical action. The delegation plays out in two phases. In *planning*, the holon drafts an executable process model that satisfies every identified requirement—assembled where possible from compliant design patterns, with each design decision made transparent—and replans when requirements change; the plan may stay abstract, deferring freedom to execution time. In *execution*, the agent binds the plan to current context (scheduling, resource assignment), passes the simulation gate before anything physical, and monitors its planning assumptions continuously—replanning within the granted degrees of freedom, escalating beyond them—while governance-level invariants, for example over personal data, remain in force. In the running example, the edge agent detects the drift with established, certifiable methods—an exponentially-weighted moving average on tool life, spectral

anomaly detection on the vibration spectra, an isolation forest on the per-cycle feature vector—with no LLM in the loop; the language model re-engages only on a pattern these deterministic detectors cannot classify.

Capabilities versus skills (Activity Theory). We model agent competence with a distinction borrowed from Activity Theory [Leontiev, 1978]: a *capability* is context-agnostic—what an agent can do in principle, advertised in its A2A Agent Card—whereas a *skill* is a context-dependent *operation*: a concrete, executable procedure (a script with declared preconditions, a bounded resource budget, and, in E3/E4, a sandbox profile) that realizes a capability under specific conditions. Routine work runs as such operations: deterministic, cheap to verify, and fast. When a skill’s script fails—a precondition is violated, a result falls out of range—the system *de-automatizes*: control rises from the operation level back to the action level, and the agent re-engages its LLM to deliberate, re-plan, or escalate to a human. De-automatization is a policy-gated transition, not a reflex: the failure is first classified, only failure classes for which the workflow pre-authorizes deliberation trigger the switch, and the reasoning path never inherits more permissions, a wider effect scope, or weaker approval requirements than the operation it replaces. Novelty, not routine, is what invokes the LLM.

Skills are derived, not hand-coded. Because operations are explicit and auditable, they can be *learned*. A shared *aggregator* reads across every agent’s private episode *vault* (the subjective record, Section 4) together with the matching telemetry (the objective record), mines recurring action patterns, and distills them into candidate skills in the shared skill repository. A skill advances through *draft* → *active* → *deprecated*, ranked by a confidence measure that rewards frequency, success rate, and cross-agent diversity—and that is deliberately downgraded when the subjective and objective records disagree (an agent claiming success over failing tool spans). Confidence ranks candidates; it never releases them: promotion to *active* passes the tests and reviews that the skill’s risk class and target environment zone require, so no skill activates itself on the strength of its own runs. In the running example, the aggregator distills the spindle-drift/forging-blank correlation into a new skill (a tightened incoming-hardness check). Because it refines *how* an authorized inspection runs rather than widening the envelope, it passes the routine test gate without a fresh safety sign-off; one that altered the envelope would need both.

Self-improvement is bounded. This learning loop is what lets a deployed agent improve over time—but it is gated, not autonomous. A skill whose risk relevance is *envelope-extending* or *envelope-altering* can never auto-promote to *active*; it is held in *pending-risk-review* until a safety engineer signs a clause in the agent’s instance-level risk assessment (Section 4). Self-improvement thus operates strictly inside the operator-signed safety envelope: agents may refine *how* they perform an authorized operation, but they cannot unilaterally widen *what* they are allowed to do. This is the reflexive L4 loop of Section 2, made safe—and made measurable: growing autonomy is a target variable, not a blank check, so progress is tracked across task success,

task horizon, interventions, cost, recovery, and policy violations, never inferred from a single autonomy score. The same episode transparency feeds one more reflexive loop above skills: large process models [Kampik *et al.*, 2024] mine the accumulated episodes and telemetry for bottlenecks and deviations and propose improvements to the *processes* the agents execute—proposals that enter operation through exactly the same gated path.

6 Interaction: Among Agents and With Humans

No agent acts alone. An industrial agent is embedded in two relationships at once—with the other agents it must coordinate with, and with the humans who remain accountable for it—and both are governed by explicit contracts rather than by shared code. We take them in turn.

Among agents. Industrial work is rarely the job of a single agent. A neuro-symbolic agent coordinates with others—inside its organization and, increasingly, across enterprise boundaries—whenever a task exceeds its authority or its capability. Here, the MAS legacy of Section 3.1 is directly useful: the negotiation primitives of FIPA-ACL and the Contract Net Protocol, and the holonic decomposition of manufacturing control, are patterns that today’s coordination standards (A2A for discovery, CloudEvents for transport) carry but do not themselves prescribe. Because participants are heterogeneous (Section 2)—rule-based controllers, LLM-driven planners, and legacy software agents speaking FIPA-ACL or the Industry-4.0 language—interaction must be defined at the level of messages and contracts, never shared code. The interaction protocols are themselves contracts: they fix not only the admissible message sequences but also which data may—and which must not—cross the organizational boundary, so compliance and data-sovereignty constraints are encoded in the protocol rather than left to each implementation. Partner selection is contract-driven too: agents advertise harmonized self-descriptions (A2A Agent Cards anchored in their AAS), enabling capability- and compliance-based matching before negotiation starts; and negotiation strategies are learnable artifacts that, like skills, are refined from episode evidence at L4 without ever bypassing the gates of Section 4. In the running example, the supply-chain agent issues a machine-interpretable quality claim to the supplier’s agent across the company boundary, and a days-long reclamation loop collapses to seconds.

With humans. The human is not a fallback but a designated role. Three channels matter. First, *authority*: a safety engineer signs the risk assessment and an operations manager the operations envelope (Section 4); the human decides *what* an agent may do, the agent only *how*. Second, *control transfer*: when an operation de-automatizes or an agent’s uncertainty crosses a threshold, the agent escalates—surfacing its intention, the options it weighed, and the residual uncertainty—so a human can intervene at the right level, from approving a single action to taking over entirely. Third, *instruction*: every human correction—an amended plan, a rejected proposal, an edited process model—is recorded in the episode and treated

as a learning signal, so oversight does not merely gate the agent but steers what and how it learns. Beyond oversight, humans and agents also form *mixed teams* that share a task rather than hand it off: on the shop floor, a worker and a neuro-symbolic agent cooperate inside a human–robot collaboration cell (E4), where the agent’s moves pass the simulation gate of Section 4 before they reach the shared physical space; and across enterprise boundaries, human decision-makers and agents from different companies drive one make-to-order process to completion. Because the same contracts govern agent-to-agent and human-to-agent interaction, a person is simply another participant—able to occupy any role an agent could. This mixed-initiative stance, with the human as decision-maker and the agent as a transparent proposer, is what makes higher-autonomy agents acceptable in E3 and E4 at all.

7 Contribution to the Trustworthy Agentic AI Roadmap

We propose five concrete items for the Trustworthy Agentic AI Roadmap [Maybury *et al.*, 2026], each in the form *gap* → *capability* → *outcome* → *time frame* → *roadblock*.

Item 1 (2026). *Gap:* no normative AAS submodel for agentic actors. *Capability:* a protocol-layer specification fixing the verifiable contracts. *Outcome:* a reference specification (proposed submodels `AIAgentType` and `AIAgentInstance`) for community review. *Roadblock:* bandwidth in the IDTA submodel working group.

Item 2 (2026–27). *Gap:* no machine-interpretable risk-assessment binding for agents. *Capability:* IDTA submodel proposals `AIAgentType` / `AIAgentInstance` carrying the `RiskAssessment` binding of Section 4—and, by the same pattern, further policy bindings (quality, data governance). *Outcome:* open-review IDTA submodel drafts. *Roadblock:* alignment between IDTA governance and EU AI Act harmonized standards.

Item 3 (2027). *Gap:* no industrial-grade reference implementation. *Capability:* a reference implementation plus two industrial pilots instantiating two of the use cases of Section 2—one in E3 (GMP-regulated pharmaceutical production) and one in E4 (an HRC cell, or an AGV fleet, under $SIL \geq 2$)—together with a machine-checkable conformance suite (versioned profiles with reference and negative fixtures), so that “implements the substrate” becomes a testable claim. *Outcome:* validated reference deployments with documented audit trails, targeting technology readiness level (TRL) 5 at project end and reference-customer pilots at TRL 6–7. *Roadblock:* pilot-partner safety-engineering capacity.

Item 4 (2028). *Gap:* governance frameworks are not interoperable at runtime. *Capability:* a published cross-mapping between EU AI Act Article 9, NIST AI RMF, ISO/IEC 42001 and ISO/IEC 23894, expressed as machine-interpretable claims an agent can carry. *Outcome:* agents that simultaneously present evidence of compliance under multiple regimes. *Roadblock:* divergence in evidence-of-conformity expectations across jurisdictions.

Item 5 (2028–29). *Gap:* agent capability markets are hyperscaler-locked. *Capability:* sovereign agent and skill marketplaces operating on open agent protocols, allowing operators to source third-party skills without surrendering data or runtime to a single vendor. *Outcome:* at least two operator-controlled skill repositories in productive use. *Roadblock:* economic incentives currently favor closed ecosystems.

Cross-cutting roadblocks include AAS-tooling maturity, key governance across supply chains (especially for the three-key authority model of Section 4), and auditor acceptance of LLM-supported decisions even when those decisions sit at L3 or L4.

8 Conclusion

The shop floor will not adopt agentic AI until safety—and compliance at large—sovereignty, and auditability stop being slogans and start being machine-checkable contracts. We have argued that today’s three relevant worlds—the forgotten MAS legacy, the agile agent-standards stack, the mature industrial-standards stack—each contribute essential pieces, but none alone suffices. We have proposed a protocol-first substrate that bridges them: neuro-symbolic agents that delegate to verifiable methods and simulation, bound risk assessments, gated skill promotion, hash-chained episodes, separated authority, and disciplined interaction among agents and with humans. Sovereignty follows by construction: each agent’s append-only episode record stays on premises under operator control, the LLM provider remains an exchangeable external dependency, and the AAS acts as a *second witness* that anchors identity and risk binding in the operator’s own asset lifecycle rather than a vendor account—so the air-gapped line of our running example runs without internet, yet still issues its cross-company quality claim over A2A with no operational data leaving operator control. Such a substrate does not automate compliance but makes it *verifiable*: the hazard analysis itself stays the responsibility of the operator and the certified safety engineer. The roadmap items above translate this proposal into a concrete sequence of community deliverables for 2026–2029, and we invite the SAFER community to shape the substrate together.

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